SMARTVLA: A STEERABLE ACTION INTERFACE FOR FINE-GRAINED, MULTIMODAL, AND CORRECTIVE INSTRUCTIONS

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ABSTRACT

Existing vision-language-action (VLA) models can perform diverse manipulation tasks from task-level instructions, but strong task performance does not ensure fine-grained semantic following, multimodal instruction understanding, or mid-execution corrective control. To this end, we propose SmartVLA, which constructs semantic and spatial supervision through action-aligned segmentation and grade-conditioned instruction augmentation, and combines warmup, pretraining, and adaptation to explicitly learn the mapping from current instructions to local actions. The resulting unified instruction interface supports natural human steering and enables SmartVLA to serve as a low-level control interface for higher-level agentic systems. To systematically assess these three capabilities, we introduce LIBERO-SMART, a benchmark comprising Semantic, Visual Prompt, and Control suites with 120 tasks and 6,000 evaluation instances. We also evaluate basic task performance, generalization across distributions, and real-world deployment on LIBERO, LIBERO-Plus, LIBERO-Pro, RoboTwin, and the Piper real-robot platform. Experiments show that SmartVLA preserves strong task performance while more reliably following diverse instructions, visually specified commands, and instructions updated during execution.

1 INTRODUCTION

Vision-language-action (VLA) models (Yu et al., 2026; Liu et al., 2026; Xiaomi Robotics Team et al., 2026; Zhang et al., 2026; Ma et al., 2026; Wu et al., 2026; Jiang et al., 2026; Generalist Team, 2026a;b; Genesis AI, 2026) combine pretrained vision-language representations with action generation and have achieved strong results in simulated and real-world manipulation. On many benchmarks, VLA success rates have approached 98%, seemingly suggesting that these models can act on human instructions flawlessly.

However, when evaluation moves beyond standard benchmarks toward more **open-ended environments**, VLA policies often prove **insufficiently SMART** in following real-world instructions. A counterintuitive example illustrates this limitation. A policy may successfully execute "pick up the cup and place it on the plate," an instruction closely resembling those in its training data, yet fail to follow the simpler **fine-grained command** "approach the cup" or even "move forward," instead producing unrelated movements or falling back on the familiar pick-and-place routine. Similar failures arise when users convey intent through **multimodal instructions**: given a trajectory drawn on the image or a red circle marking the object to grasp, the policy often disregards the cue and moves toward a familiar target. Likewise, when a robot makes an error or is about to perform an unsafe action, a human operator may issue **corrective commands** such as "stop" or "go around to the right, not the left," yet the policy frequently continues along its original course. These cases reveal a substantial gap between current VLA capabilities and genuine instruction-following ability.

To quantify this gap, we conduct controlled behavioral interventions on three representative VLAs, $\pi_{0.5}$ (Physical Intelligence et al., 2025b), GR00T N1.6 (GEAR Team, 2025), and StarVLA (StarVLA Community, 2026), using LIBERO-based manipulation tasks. As shown in Figure 1, all three models achieve 96–99% complete-task success with the original task instructions and scenes. When instructions instead specify fine-grained actions, success drops sharply to 1–6%.

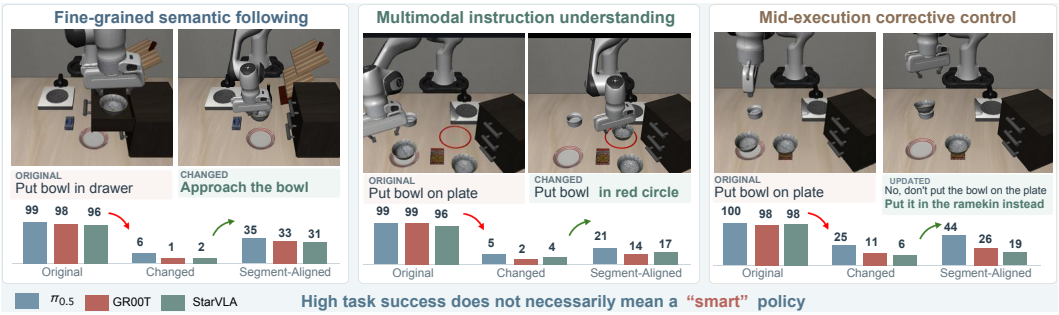


Figure 1: Instruction-following diagnosis across fine-grained semantic following, multimodal instruction understanding, and mid-execution corrective control. Bars compare success rates (%) under original instructions, changed or updated instructions, and segment-aligned supervision for $\pi_{0.5}$, GR00T, and StarVLA. High original-task success can coexist with poor instruction following, while segment-aligned supervision improves performance under the changed requirements.

When explicit visual prompts specify the object or spatial target, success is only 2–5%. For corrective control, we initialize the robot in an intermediate state with an object already held and update the target or subsequent execution requirement; success averages only 14% across the three models, ranging from 6% to 25%. These results show that even models with strong manipulation benchmark performance remain far from sufficiently smart instruction following.

We summarize these failures as three limitations in instruction following: ① **fine-grained semantic following**, where policies fail to execute the local action, scope, or constraint specified by an instruction and instead default to familiar complete tasks; ② **multimodal instruction understanding**, where policies fail to ground explicit visual prompts in the intended object, target, or motion; and ③ **mid-execution corrective control**, where policies fail to revise subsequent actions once the instruction is updated during execution. Without these capabilities, current VLA policies are not smart enough to serve as reliable low-level controllers, hindering the translation of high-level plans and human corrections into flexible, responsive behavior in the physical world.

A plausible explanation is that task-level supervision pairs a single instruction with an entire action sequence but provides little explicit guidance on which parts of the instruction govern each local action. Under such supervision, a policy may rely on visual cues in familiar scenes to reproduce action sequences without adequately using the current instruction to determine the required local actions, thereby taking visual shortcuts that bypass instruction requirements. To examine whether explicitly aligning local instructions with action segments improves instruction following, we divide demonstration trajectories into K consecutive local action segments based on action phases and annotate each segment with its corresponding local instruction. As shown in Figure 2, in a small-scale training comparison, we evaluate two supervision strategies: concatenating all local instructions to supervise the entire action sequence yields success rates of **23%, 16%, and 32%** in the fine-grained instruction, visual-prompt, and corrective-control settings, respectively; pairing each local instruction with its corresponding action segment during training further improves these rates to **35%, 21%, and 44%**. These results indicate that richer instruction annotations improve instruction following, while explicitly aligning local instructions with their corresponding action segments provides additional gains beyond language enrichment alone.

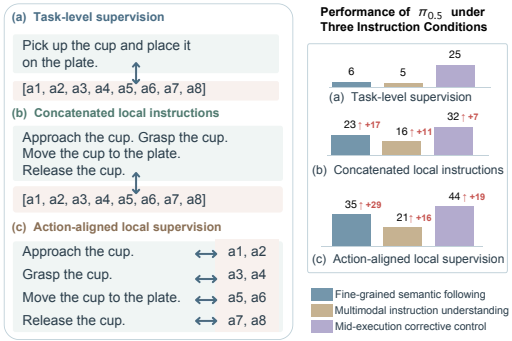


Figure 2: Effect of supervision granularity on $\pi_{0.5}$ instruction following.

We therefore propose SmartVLA, a framework centered on supervision that aligns local instructions with action segments to improve fine-grained semantic following, multimodal instruction understanding, and mid-execution corrective control in VLA policies. Motivated by the preceding

experiment on supervision granularity, we explicitly pair local instructions with action segments to provide fine-grained supervision, rather than relying solely on a single task-level instruction for an entire demonstration. To construct this segment-level supervision, we develop an automated action-aligned data construction pipeline that first determines the scope of local behavior from action and interaction evidence in demonstrations, then generates and reviews the corresponding local instructions. By grounding instruction generation in demonstrated behavior, the pipeline reduces reliance on costly end-to-end video annotation and per-segment manual rewriting. To further improve multimodal instruction understanding, the pipeline also adds visual prompts to the image observations of applicable segments to specify objects, spatial targets, or motion paths. These prompts are paired with the corresponding local instructions and action segments to form multimodal training examples.

However, even when local instructions are aligned with action segments, a policy may still rely excessively on visual cues from familiar scenes without sufficiently using the current instruction to generate actions. We therefore adopt **three training stages: warmup, pretraining, and adaptation**. Warmup learns the correspondence between instructions and motion without image inputs; pretraining restores visual inputs to learn how to execute instructions in the current scene; and adaptation supports both complete-task execution and local instruction following in the target domain. The policy can accept updated instructions in subsequent calls, allowing human operators or upper-level models to steer ongoing execution.

To evaluate whether these capabilities are realized in existing policies, we introduce LIBERO-SMART, which separately assesses three forms of instruction following: fine-grained semantic following, where the policy must execute the specified local action, relation, or constraint; multimodal instruction understanding, where visual prompts specify the current object or target location; and mid-execution corrective control, where the policy must adapt its subsequent actions after a command update or redirection. Although $\pi_{0.5}$ achieves 96.85% on the original LIBERO benchmark, its average success rate on LIBERO-SMART is only 7.18%. This gap is consistent with our diagnostic observations: a policy may possess strong basic manipulation skills while failing to follow semantic requirements and visual prompts or to redirect its behavior after an intermediate command change.

On LIBERO-SMART, SmartVLA-Pi and SmartVLA-OFT achieve average success rates of 14.67% and 14.03%, respectively, compared with 7.18% for $\pi_{0.5}$, with both variants outperforming all direct-policy baselines across the three suites. Ablation studies support the contributions of action-aligned segmentation, grade-conditioned rewriting, spatial supervision, and staged training to these gains. We also evaluate standard task performance on LIBERO, LIBERO-Plus, LIBERO-Pro, and RoboTwin, as well as real-world instruction following on the Piper platform. SmartVLA-Pi achieves 98.30% on LIBERO and 84.00% on LIBERO-Plus, outperforming $\pi_{0.5}$ by 1.45 and 3.74 percentage points, respectively. On Piper, it raises average success from 45.00% to 58.50%, a gain of 13.50 percentage points. These experiments show that SmartVLA enables smarter, more flexible, and more controllable robot behavior while maintaining strong performance on standard benchmarks.

Our contributions are:

- We develop SmartVLA to connect current instructions with local control. An action-aligned pipeline reconstructs existing demonstrations into phase-specific semantic and visual supervision, while staged training establishes a language-conditioned action prior, grounds it in observations, and adapts it to target domains with complete-task and local supervision.

- We introduce LIBERO-SMART, comprising Semantic, Visual Prompt (Vis), and Control suites with 120 tasks and 6,000 evaluation instances across diverse initializations, to evaluate fine-grained semantic following, multimodal instruction understanding, and mid-execution corrective control.

- We systematically evaluate standard task performance and the three instruction-following capabilities across two VLA architectures on the LIBERO benchmarks, RoboTwin Full and C2R, LIBERO-SMART, and the Piper real-robot platform. We further assess callability under a fixed upper-level protocol.

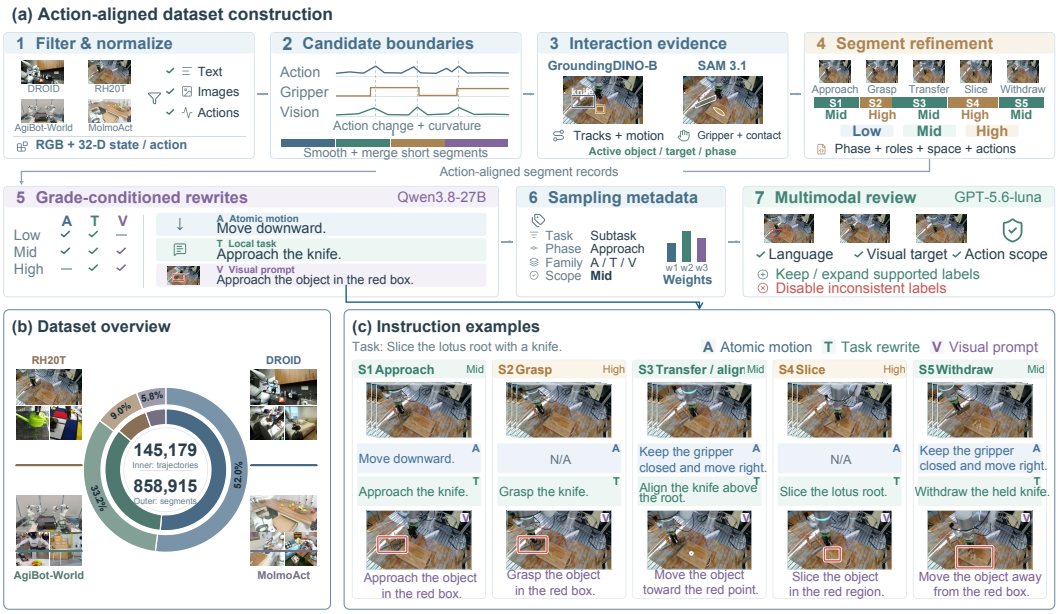


Figure 3: Action-aligned data construction and supervision. (a) Seven-stage pipeline from raw demonstrations to reviewed segment annotations. (b) Source composition of the retained trajectories (inner ring) and action segments (outer ring). (c) Grade-conditioned atomic-motion (A), task-language (T), and visual-prompt (V) examples from an RH20T demonstration; N/A denotes an inapplicable rewrite family.

2 SmartVLA Framework

SmartVLA builds on pretrained representations to strengthen instruction-conditioned control while retaining complete-task execution. Our core idea is to refine the granularity of instruction–action supervision by explicitly pairing local instructions with their corresponding action segments while retaining the original task-level supervision. For data construction, we use automated annotation to establish these instruction–action pairs for supervision (Section 2.1). For training, we adopt a staged procedure to encourage the policy to learn and use this correspondence while preserving complete-task supervision (Section 2.2).

2.1 Action-Aligned Data Construction Pipeline

We represent local supervision as

$$r_i = (\mathbf{O}_i, \mathbf{S}_i, \mathbf{A}_i, c_i, z_i, g_i),$$

where observations, robot states, and actions span a segment of length L_i, while c_i, z_i, and g_i denote the segment's instruction, object/target evidence, and interaction grade, respectively.

To construct this segment-level supervision, we organize the conversion from raw demonstrations to training examples into seven interconnected stages. Figure 3(a) illustrates the automated annotation pipeline.

1. **Quality filtering and representation unification.** We filter trajectories from multiple sources to remove samples with missing language annotations, unusable visual observations, or anomalous states or actions. We then unify the representations of observations, robot states, and actions and record the valid dimensions for each robot configuration, providing a common interface across data sources while preserving their valid control information.
2. **Candidate boundary generation.** We generate candidate boundaries from differences between consecutive actions, motion curvature, gripper opening and closing events, and

cross-frame visual changes. DINOv3 is used to represent cross-frame visual changes. Temporal smoothing and the merging of excessively short segments reduce oversegmentation caused by transient fluctuations, yielding candidate control segments.

3. **Object and interaction evidence extraction.** We use GroundingDINO to localize candidate objects and SAM to obtain object or target-region masks. Temporal tracking associates object positions across consecutive frames. We then combine gripper–object relative positions, object motion, gripper state, and contact evidence to infer the active object, target region, and interaction phase.
4. **Segment refinement and interaction grading.** We combine candidate boundaries with interaction evidence and refine segment boundaries using temporal consistency and task-family rules. Each segment is assigned a local behavioral scope and a Low, Mid, or High interaction grade. The resulting structured records contain action segments, control phases, object roles, and spatial evidence, providing a basis for subsequent instruction generation.
5. **Grade-conditioned instruction rewriting.** We select applicable rewrite families according to the interaction grade and use Qwen to generate candidate local instructions from the original task and structured segment records. We also construct visual-prompt training examples for applicable segments. Candidate instructions and visual prompts are associated with their corresponding demonstration action segments; the applicability rules are described below and summarized in Table 1.
6. **Sampling metadata recording.** For each candidate rewrite, we record the task type, control phase, rewrite family, applicability conditions, and recommended sampling weight. This metadata supports the subsequent organization of training examples by supervision type and scope of applicability.
7. **Multimodal consistency review.** Using keyframes, candidate instructions, and spatial annotations, we check consistency among instruction meaning, visual targets, and segment behavior. We add alternative expressions where supported by the evidence and remove or disable language and spatial labels that are inconsistent with the action scope, object relations, or visual target.

This pipeline first establishes the local action scope and interaction context from demonstration evidence, then performs constrained instruction generation, without requiring the language generation model to independently infer the entire temporal structure from a complete video.

Grade-conditioned rewriting rules. In the fifth stage, interaction evidence constrains the applicable rewrite families so that instruction content remains consistent with the local behavioral scope of the corresponding segment. We use three rewrite families: Object-free atomic motion, which describes local motion using instructions without object names; Synonym/task rewrite, which varies the wording while preserving the local goal and execution scope; and Spatial visual generalization, which pairs visual prompts such as points, boxes, regions, or trajectories in the observation images with the corresponding local instructions and action segments. Visual prompts are provided to the policy as part of the image observations, while action supervision comes from the same demonstration segment.

For different phases of a task, we define rewriting strategies conditioned on three grades: Low (g_0), Mid (g_1), and High (g_2). These grades correspond to free-space motion, transitions, and explicit object interaction, respectively. Low permits only atomic-motion and synonym/task rewrites, avoiding spatial references when no clear interaction target is available; Mid permits all three rewrite families; and High permits only synonym/task rewrites and spatial visual generalization, avoiding the reduction of explicit object interactions to atomic motions that omit essential object relations. For example, "move up" does not require identifying an object, whereas a placement instruction must retain information about both the object to be placed and its target location. Interaction grades therefore govern rewrite applicability and specificity, rather than serving as a criterion for segment filtering. Figure 3(c) illustrates an example.

Table 1: Grade-conditioned rewrite families.

Grade	Atomic motion	Synonym/ task	Spatial visual
Low	yes	yes	no
Mid	yes	yes	yes
High	no	yes	yes

Final supervision format. Through this pipeline, each local instruction is associated with the corresponding observations, robot states, and demonstration action segment, forming the local super-

vision dataset $\mathcal{D}_{\text{local}}$. The same action segment can yield multiple training examples through different applicable rewrites, while its action supervision continues to come from the corresponding demonstration. We also retain complete trajectories and their original task instructions to form the task-level supervision dataset $\mathcal{D}_{\text{task}}$, providing complementary supervision for local instruction following and complete-task execution. Data composition and segmentation statistics are shown in Figure 3; implementation details and segment-level examples are provided in Appendix D.

2.2 Three-Stage Training Recipe

We separate learning how instructions specify motion, how observations locate its target, and how both transfer to a target domain. Let $\theta = (\psi, \phi)$ collect the backbone and action-head parameters. The policy predicts $\mathbf{A}_t = \mathbf{a}_{t:t+H-1} \sim \pi_\theta(\cdot \mid \mathbf{o}_t, \mathbf{s}_t, c_t)$; prediction length H is distinct from segment length L_i. All stages retain the native per-example action loss ℓ_{act}, with objective

$$\mathcal{L}_{\text{act}}(\theta; \mathcal{D}) = \mathbb{E}_{(\mathbf{o},\mathbf{s},c,\mathbf{A})\sim\mathcal{D}} \left[\ell_{\text{act}}(\theta; \mathbf{o}, \mathbf{s}, c, \mathbf{A})\right]. \tag{1}$$

Warmup: learn motion correspondence without image-based shortcuts. Atomic instructions specify motion without requiring object localization. Warmup therefore receives only atomic instructions and robot state, freezes the backbone at ψ_0, and trains the action head with $\mathcal{L}_{\text{w}}(\phi) = \mathcal{L}_{\text{act}}((\psi_0, \phi); \mathcal{D}_{\text{atomic}})$. Here $\mathcal{D}_{\text{atomic}}$ contains eligible Low/Mid segments with no image input. Different atomic commands specify corresponding motion patterns, while robot state supplies the physical conditions for execution. Removing images prevents initial learning from directly relying on visual scene cues; freezing the backbone focuses optimization on how the action head uses existing representations, establishing a language-conditioned action prior for subsequent visually conditioned learning.

Pretraining: ground the prior in current scenes. Motion alone does not identify its target: "lift the object in the red box" also requires binding the prompt to the current object. Starting from warmup, we restore multi-view images and train both modules with $\mathcal{L}_{\text{p}}(\theta) = \mathcal{L}_{\text{act}}(\theta; \mathcal{D}^{\text{pre}}_{\text{mix}})$. The mixture $\mathcal{D}_{\text{mix}} = \alpha\mathcal{D}_{\text{task}} + (1-\alpha)\mathcal{D}_{\text{local}}$ samples original-task and local supervision, respectively. Local commands and visual prompts teach scene-dependent control, while original-task examples retain complete-task supervision. The resulting checkpoint is SmartVLA-Base.

Adaptation: retain both supervision granularities. Original-task instructions alone do not preserve local command–action supervision, while local segments cannot replace supervision of complete execution. We therefore reconstruct target-domain demonstrations using Section 2.1 and independently adapt each domain d from SmartVLA-Base, optimizing both modules with $\mathcal{L}^d_{\text{a}}(\theta) = \mathcal{L}_{\text{act}}(\theta; \mathcal{D}^d_{\text{mix}})$. This continues local language and visual supervision while learning target-domain control.

SmartVLA-Pi and SmartVLA-OFT follow this recipe with their native action models. Both accept complete-task, subtask, or updated corrective commands through the same observation-conditioned interface. Architectures, initialization, sampling ratios, training settings, and the calling protocol are detailed in Appendix E.

3 LIBERO-SMART

LIBERO-Pro, LIBERO-Plus, and RoboTwin-IF evaluate generalization across changes in task goals, layouts, observations, and language requirements. Such variations are important, but robust task completion does not by itself establish that a policy follows the precise scope and execution conditions of a current instruction. For example, rephrasing "pick up the cup and place it on the plate" preserves the intended grasp-and-place sequence, whereas "only approach the cup" requires suppressing its later stages. LIBERO-SMART makes these distinctions explicit through Semantic, Visual Prompt, and Control tasks and instruction-consistent trajectory contracts (Figure 4).

3.1 Instruction Generalization over Familiar Skills

The benchmark asks whether familiar manipulation capabilities can be invoked under new requirements, rather than requiring a new skill for every command. A policy that has learned grasping, transport, and placement should be able to select the requested phase, bind its target to the current

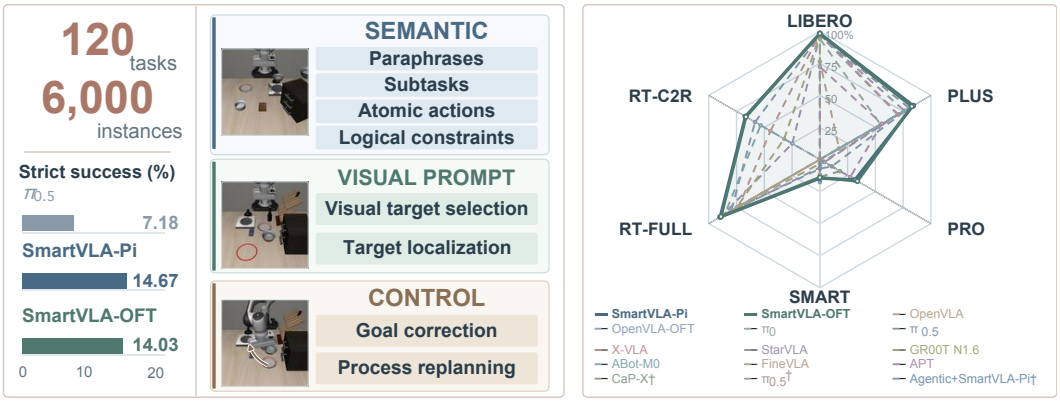


Figure 4: LIBERO-SMART overview and performance comparison. Left: the benchmark comprises Semantic, Visual Prompt, and Control suites, with 120 tasks and 6,000 evaluation instances; bars report average strict success rates on LIBERO-SMART. Right: success rates across the LIBERO series and RoboTwin Full and C2R benchmarks.

observation, or redirect its continuation. We therefore retain some LIBERO scene assets and basic skills while evaluating held-out skill combinations, language conditions, object–target relations, and layouts. This separates familiar motor operations from the new conditions governing their use.

Different commands alone cannot exclude solutions based on familiar object locations. Each task therefore has 10 substantially different macro layouts and 5 local position and pose variants per layout, preserving instruction meaning and physical feasibility while changing object configurations and target locations. For Control, models start from the same intermediate physical state within each instance, so differences in preceding-stage execution do not change the difficulty of the corrective continuation.

Reaching a target also does not establish that the full instruction was followed: a policy may first manipulate a forbidden object, execute steps in the wrong order, or immediately undo the intended result. Task contracts therefore jointly assess target correctness, process constraints, and result stability, including applicable prohibitions, step-order constraints, and release or holding requirements. Together, tasks, initializations, and contracts test whether improved instruction–action correspondence yields complete instruction execution. Training–evaluation separation and detailed construction and scoring rules are provided in Appendix F.

3.2 Three Task Suites

LIBERO-SMART comprises three independently scored suites, Semantic, Visual Prompt (Vis), and Control, with 40 tasks per suite and 50 initializations per task, totaling 6,000 evaluation instances. Both direct policies and agentic systems are evaluated using the same trajectory-level task contracts.

Semantic. This suite evaluates fine-grained semantic following through requirements on the manipulated object, action scope, and termination condition. It covers equivalent semantic rewrites, designated subtasks of a complete episode, standalone atomic actions, and logical constraints on object exclusivity, spatial relations, pose, and order. For example, when an instruction only requests picking up an object, subsequently placing it in the original task container violates the requirement.

Visual Prompt. This suite evaluates multimodal instruction understanding by testing whether models can bind explicit image cues to the current object or spatial target. Cues include circles identifying objects to manipulate, circles defining placement regions, points specifying target locations, and trajectories guiding motion. They refer to the current object or location in each initialization and remain consistent with the language instruction and target evaluation.

Control. Starting from an intermediate physical state, this suite evaluates mid-execution corrective control by testing whether models adjust subsequent behavior according to a new instruction. Target correction changes the intended final result, for example redirecting a held object to another tar-

get. Process re-planning changes how subsequent actions are executed, for example holding stably, reversing an operation, or retaining completed parts while rearranging the remaining steps.

4 EXPERIMENTS

Unless otherwise specified, SmartVLA results are reported using the corresponding target-domain-adapted experts. For LIBERO, LIBERO-Plus, LIBERO-Pro, and RoboTwin2.0, we follow the evaluation settings released with each benchmark. LIBERO-SMART task contracts and scoring are specified in Appendix F; the Piper real-robot protocol is described in Appendix I. Figure 4 summarizes LIBERO-SMART and compares performance across simulation benchmarks. All success rates are reported as percentages.

4.1 LIBERO-SMART

Table 2: LIBERO-SMART success rates (%). † denotes an agentic system.

Method	Sem	Vis	Con	Avg.	Goal-only
OpenVLA-OFT	1.45	0.30	1.15	0.97	1.65
π_0	0.00	0.00	0.00	0.00	0.85
$\pi_{0.5}$	14.70	2.00	4.85	7.18	17.92
X-VLA	0.00	0.00	0.00	0.00	0.55
GR00T N1.6	8.35	0.80	1.00	3.38	13.22
StarVLA	12.60	1.95	4.00	6.18	15.62
CaP-X†	12.75	6.75	**21.25**	13.58	18.33
Agentic+$\pi^{\dagger}_{0.5}$	17.75	3.50	14.25	11.83	18.22
SmartVLA-Pi	23.05	**8.40**	12.55	14.67	29.73
SmartVLA-OFT	20.10	5.55	16.45	14.03	32.73
Agentic+SmartVLA-Pi†	**26.50**	7.00	19.50	**17.67**	**37.97**

Table 2 evaluates whether SmartVLA improves the three instruction-following capabilities targeted by our method. In direct-policy evaluation, SmartVLA-Pi and SmartVLA-OFT achieve average success rates of 14.67% and 14.03%, compared with 7.18% for $\pi_{0.5}$, with improvements across all three suites. The Semantic gains indicate better execution of local commands and instruction-specific constraints; the Visual Prompt gains reflect improved use of visually specified objects and target locations; and the Control gains show better adjustment to commands issued during execution. These improvements are consistent with learning segment-aligned commands and visual prompts as conditions for subsequent actions. Under Goal-only scoring, which checks any-time goal attainment without the additional contract requirements, the two variants also improve from 17.92% for $\pi_{0.5}$ to 29.73% and 32.73%. Their advantage therefore extends beyond the strict scoring criterion.

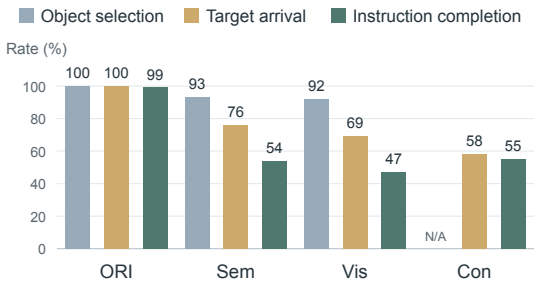


Figure 5: SmartVLA-Pi behavioral diagnosis (%).

Behavioral diagnosis. Figure 5 revisits the motivating interventions with SmartVLA-Pi. Instruction-completion rates increase from 6.00%, 5.00%, and 25.00% for $\pi_{0.5}$ to 54.00%, 47.00%, and 55.00% on Semantic, Visual Prompt, and Control, respectively, while original-task completion remains at 99.00%. This comparison connects the benchmark improvements to more reliable execution of local instructions, visual prompts, and command updates.

Under a fixed agentic protocol, replacing $\pi_{0.5}$ with SmartVLA-Pi raises strict system success from 11.83% to 17.67% and Goal-only success from 18.22% to 37.97%. Thus, improved low-level in-

struction following also benefits systems that invoke the policy through an upper-level controller. Calling and scoring protocols are detailed in Appendices E.5 and F.3.

4.2 Generalization on Simulation Benchmarks

Table 3: Success rates on LIBERO, LIBERO-Plus, and LIBERO-Pro. † denotes an agentic system.

Method	LIBERO	LIBERO-Plus	Pro-Pos.	Pro-Task	Pro-Avg.
OpenVLA	76.50	17.28	0.00	0.00	0.00
OpenVLA-OFT	97.10	70.02	5.75	0.63	3.19
π_0	94.15	54.63	0.00	0.00	0.00
$\pi_{0.5}$	96.85	80.26	20.75	0.75	10.75
X-VLA	98.05	70.46	0.75	6.75	3.75
StarVLA	96.60	76.08	4.30	3.90	4.10
APT	72.25	55.85	27.00	27.75	27.38
CAP-X†	–	–	20.00	16.33	18.17
SmartVLA-Pi	**98.30**	**84.00**	30.95	32.35	31.65
SmartVLA-OFT	98.20	82.76	**31.50**	**36.10**	**33.80**

LIBERO, LIBERO-Plus, and LIBERO-Pro. Table 3 examines task performance and generalization on benchmarks developed independently of LIBERO-SMART: LIBERO (Liu et al., 2023), LIBERO-Plus (Fei et al., 2025), and LIBERO-Pro (Zhou et al., 2025). SmartVLA-Pi and SmartVLA-OFT achieve 98.30% and 98.20% on LIBERO, exceeding X-VLA (Zheng et al., 2025) (98.05%) while retaining strong basic manipulation performance. On LIBERO-Plus, they reach 84.00% and 82.76%, above $\pi_{0.5}$ (80.26%) and APT (55.85%). On LIBERO-Pro, both variants outperform APT on Position and Task, reaching averages of 31.65% and 33.80% compared with 27.38%, while also substantially exceeding APT's original LIBERO performance of 72.25%. These results show improved robustness to benchmark perturbations and changes in task goals and scene layouts without a loss of strong original-task performance. Detailed results are provided in Appendix G.

Table 4: RoboTwin2.0 Full and Clean2Random results. C/R denotes clean/randomized test scenes.

Method	Full-C	Full-R	Full Avg.	C2R-C	C2R-R	C2R Avg.
$\pi_{0.5}$	82.70	76.80	79.75	70.70	46.00	58.35
X-VLA	72.80	72.84	72.82	68.00	20.90	44.45
GR00T N1.6	74.76	75.08	74.92	43.60	20.70	32.15
StarVLA	88.18	88.32	88.25	46.50	3.20	24.85
ABot-M0	86.10	85.10	85.60	70.70	36.00	53.35
FineVLA	86.80	82.50	84.65	–	–	–
SmartVLA-Pi	**89.86**	**89.72**	**89.79**	**75.70**	58.40	**67.05**
SmartVLA-OFT	88.94	89.06	89.00	74.02	**59.08**	66.55

RoboTwin2.0. Table 4 extends the evaluation to bimanual manipulation on RoboTwin2.0 (Mu et al., 2025; Chen et al., 2025). Under Full, which includes randomized scenes during adaptation, SmartVLA-Pi and SmartVLA-OFT achieve average success rates of 89.79% and 89.00%, exceeding StarVLA (88.25%) and ABot-M0 (Yang et al., 2026) (85.60%). Under Clean2Random, adaptation uses only clean demonstrations, yet the two variants reach averages of 67.05% and 66.55%, compared with 58.35% for $\pi_{0.5}$. Their success rates on randomized test scenes reach 58.40% and 59.08%, respectively, versus 46.00% for $\pi_{0.5}$, demonstrating stronger transfer to unseen scene configurations.

Together, these results show that the benefits are not confined to a benchmark explicitly designed around our three instruction-following capabilities. Improvements on independently developed task and generalization benchmarks support the broader value of aligning language and visual supervision with local actions.

4.3 Ablation Studies

Table 5 examines the contributions of training stages and supervision construction. Removing warmup or pretraining, or adapting directly from the original VLM, reduces performance, supporting the three-stage procedure. Adaptation using only original demonstrations also degrades performance, indicating that local instruction supervision remains important in the target domain. Uniform temporal segmentation reduces average success from 14.67% to 7.45%, supporting alignment with control phases rather than merely annotating shorter trajectories. Removing grade conditioning further weakens performance. The FineVLA-inspired fine-grained semantic-only setting reaches 7.00%, indicating that enriching whole-trajectory language alone does not recover the benefits of the full construction. These comparisons support the roles of action-aligned segmentation, phase-appropriate rewriting, and staged training; setting definitions are provided in Appendix H.

Table 5: SmartVLA-Pi ablations on LIBERO-SMART (%).

Setting	Sem	Vis	Con	Avg.
w/o warmup	19.70	7.35	11.15	12.73
w/o pretraining	13.90	2.55	5.10	7.18
From raw VLM	14.05	2.30	4.80	7.05
Raw-data adaptation	16.85	4.25	5.10	8.73
Uniform segmentation	12.20	5.55	4.60	7.45
w/o grade conditioning	20.40	6.30	10.05	12.25
Fine-grained sem. only	15.10	2.05	3.85	7.00
Full SmartVLA-Pi	**23.05**	**8.40**	**12.55**	**14.67**

4.4 Real-Robot Piper Evaluation

Beyond simulation, we evaluate fine-grained semantic following, multimodal instruction understanding, and mid-execution corrective control on a bimanual Piper platform. Each category contains 10 tasks, with 20 trials per task, for a total of 600 trials per model. The tasks build on basic operations such as grasping, placing, moving, and holding, varying language requirements, visual targets, or commands during execution to test whether policies adjust their behavior to the current instruction. Models use matched tasks, initial conditions, and interaction budgets, with results reported separately for the three capabilities in Figure 6. Representative execution trajectories are provided in Appendix I.

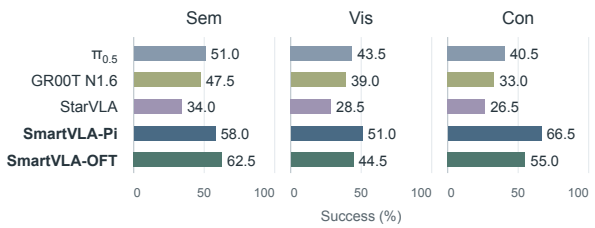


Figure 6: Real-robot instruction following on Piper (%).

In the bimanual Piper evaluation, SmartVLA-Pi and SmartVLA-OFT achieve average success rates of 58.50% and 54.00%, respectively, exceeding the strongest baseline, $\pi_{0.5}$ (45.00%), by 13.50 and 9.00 percentage points. On Semantic, Visual Prompt, and Control, the best result across the two variants exceeds $\pi_{0.5}$ by 11.50, 7.50, and 26.00 percentage points, respectively. These results show that SmartVLA also improves fine-grained semantic following, multimodal instruction understanding, and mid-execution corrective control in real-robot manipulation, rather than only improving performance on simulation benchmarks.

5 Conclusion

We present SmartVLA, a framework for training and evaluating VLA policies beyond task-level control. Behavioral and representation diagnostics reveal that decodable target information does not necessarily lead to actions consistent with the current instruction. To address this gap, SmartVLA combines action-aligned semantic and spatial supervision with three-stage training, improving fine-grained semantic following, multimodal instruction understanding, and mid-execution corrective control. We further introduce LIBERO-SMART, which uses diverse initializations and trajectory-level task contracts to assess complete instruction following rather than target attainment alone. Simulation and real-robot experiments with two policy architectures show that SmartVLA improves instruction following and generalization while maintaining strong basic task performance. These

results support fine-grained instruction supervision aligned with action segments as an effective approach to improving VLA steerability, enabling the same policy to support natural human-robot interaction and serve as a flexibly callable control interface in hierarchical robotic systems.

AI USE STATEMENT

We used AI tools in the dataset-construction pipeline to propose candidate segment-level annotations and language/spatial rewrites under predefined rules, review their semantic and visual consistency, and filter the resulting records. The authors remain responsible for the final dataset and its use. We also used language-model assistance for translation, grammar editing, and document formatting. AI was not used to generate experimental results, determine evaluation outcomes, or make the final scientific claims; all technical claims, methods, experimental settings, numerical results, and interpretations are the responsibility of the authors and have been independently verified by the authors.

REPRODUCIBILITY STATEMENT

To support reproducibility, we will release after publication the source code for data processing, annotation, training, inference, and evaluation, together with the SmartVLA model checkpoints. We will also release the LIBERO-SMART task definitions, initialization files, evaluators, associated configurations, and documentation needed to reproduce the reported experiments.

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APPENDIX

A RELATED WORK

A.1 GENERALIST VISION-LANGUAGE-ACTION POLICIES

Generalist VLA models combine the semantic understanding capabilities of vision-language models with robot control, improving task execution and generalization through advances in backbones, data scale, and action modeling. The RT family demonstrates the scalability of language-conditioned policies trained on large-scale robot data (Brohan et al., 2022; 2023; Open X-Embodiment Collaboration et al., 2023), while OpenVLA integrates an open VLM backbone with robot action prediction (Kim et al., 2024; 2025). Building on a flow-matching action expert, the π family combines heterogeneous robot data, high-level semantic prediction, and multimodal co-training to improve the open-world generalization and steerability of continuous-action policies (Black et al., 2024; Physical Intelligence et al., 2025b;a; 2026). VLA work in the Qwen family explores unified alignment of multi-source robot data at the representation, action, and behavior levels, as well as the role of data scale and alignment quality in generalization across environments, embodiments, and instructions (Wang et al., 2026; Yuan et al., 2026). However, task execution and generalization performance under task-level instructions do not directly establish whether a policy can reliably translate local commands, explicit spatial cues, and requirements updated during execution into subsequent actions. SmartVLA complements these efforts without introducing a new backbone architecture. Instead, it combines action-aligned data construction with staged training to transform existing robot trajectories into semantic and spatial supervision aligned with local control phases, learning a mapping from current instructions to local actions that supports fine-grained instruction following, responses to spatial cues, and steering during execution.

A.2 AGENTIC HARNESSES AND CALLABLE VLA INTERFACES

Recent agentic robotic systems organize execution as a closed loop linking high-level reasoning and orchestration with low-level skill invocation. Works such as CaP-X (Fu et al., 2026), ASPIRE (Lu et al., 2026), and Zetta (Ding et al., 2026) improve task decomposition, memory, tool use, and failure handling through code-based policies, skill libraries and search, and runtime critique and recovery. Stronger general-purpose multimodal models such as GPT-6 Astra (OpenAI, 2026) further advance long-horizon reasoning, visual state understanding, tool use, and dynamic replanning, enabling high-level harnesses to continuously orchestrate low-level skills based on current observations and execution feedback. However, stronger high-level reasoning and orchestration do not automatically ensure that low-level policies reliably follow updated commands and satisfy spatial requirements. SmartVLA complements these efforts: the high-level harness handles subgoal generation, skill selection, and plan revision, while SmartVLA uses the current command, visual cues, and robot state to translate semantic and spatial requirements into local actions and respond to updated instructions in subsequent calls. We compare policies under a fixed high-level harness, replacing only the low-level VLA, to assess how low-level instruction execution affects system performance without attributing gains from high-level planning, prompt engineering, or tool selection to the low-level policy.

A.3 STEERABLE AND INSTRUCTION-GENERALIZABLE POLICIES

Another line of work focuses on fine-grained language, steerability, and instruction generalization. FineVLA (Hu et al., 2026) uses fine-grained language supervision to provide execution-critical details, while Steerable Policies (Chen et al., 2026) enables low-level policies to be invoked through subtask commands. APT (Xu et al., 2026b) and BeTTER (Xu et al., 2026a) investigate issues such as sparse language supervision, visual shortcuts, vocabulary-motion associations, and behavioral inertia through action-expert pretraining and spatial-layout and temporal interventions, respectively. These works collectively emphasize that final task success rates alone cannot establish whether the current instruction actually constrains action execution. Following this direction, we focus on aligning semantic and spatial supervision with local actions, using data construction and training to improve policy responsiveness to fine-grained instructions, explicit spatial cues, and instruction updates during execution.

B Behavioral Intervention Diagnosis Protocol

B.1 Condition Construction and Instance Pairing

We evaluate $\pi_{0.5}$, GR00T N1.6, and StarVLA under four conditions: ORI, Semantic (Sem), Visual Prompt (Vis), and Control (Con). ORI retains the original task instruction and scene to measure basic task execution. Semantic keeps the scene and task context executable while changing the object, local action, target relation, or holding/termination requirement. Vis changes object or target layout, or adds a circle, point, region, or trajectory in the image as the visual target. Control starts from an intermediate state in which the robot holds the object and replaces the target or subsequent process requirement. The initial held state in Control is not counted as a new object selection.

The three models use the same instances, environment initializations, and intervention commands. Each model is evaluated on 100 instances per condition, for 1,200 rollouts in total. Semantic commands cover phase sub-tasks, atomic motions, object roles, spatial relations, and holding or termination requirements. Visual prompts are synchronized with the current object or target region, and Control commands are applied only to physically executable intermediate states. For each rollout, we record the original task, initialization, intervention command, and applicable evaluation conditions.

B.2 Behavioral Metrics

We report three binary metrics:

- **Target-object selection rate**: the specified object is correctly grasped at least once during execution. This metric is not applicable to Control, which starts with an object already held.
- **Target-position attainment rate**: the manipulated object reaches the position, region, or target relation specified by the current instruction at least once.
- **Instruction-completion rate**: all applicable relational, sequential, release, holding, and termination requirements are satisfied.

Correct object selection or transient target attainment does not imply complete execution. These metrics decompose the behavioral chain and do not replace the LIBERO-SMART trajectory contract or define a ranking across architectures.

B.3 Trajectory Review

Behavior is judged jointly from state trajectories, action trajectories, and task conditions. Representative rollouts are reviewed frame by frame to determine whether the correct object was selected, whether the target changed with the current condition, whether the obsolete task continued, and whether unnecessary actions occurred after a local goal was reached. The main recurring failure patterns are summarized in Table 6 and illustrated in Figure 7. The complete quantitative metrics are reported in Appendix C, Table 7.

B.4 Behavioral Results and Failure Patterns

All three models correctly select the object and reach the original target under ORI, with complete-task success rates of 96–99%. Under Sem, target-object selection remains at 79–88%, but target-position attainment drops to 3–11%, and instruction completion falls to only 1–6%. Correctly identifying and grasping the object therefore does not ensure that the subsequent execution satisfies the specified local action, target relation, or holding and termination requirements.

Under Vis, target-object selection remains at 72–83%, while target-position attainment is only 3–14% and instruction completion is only 2–5%. Under Con, target-position attainment and instruction completion range from 6% to 25%, despite starting from a physically executable state with the object already held. These results distinguish object selection, target-position attainment, and complete instruction execution: success at an earlier part of the behavior does not guarantee that subsequent actions satisfy the current instruction. Table 7 reports the full results.

Table 6: Recurring behavior patterns observed in trajectory review.

Pattern	Observable behavior	Capability gap
Target carryover	Correct grasp, original target retained	Target rebinding
Visual-target decoupling	Placement not bound to the visual cue	Visual-prompt grounding
Command inertia	Old sequence continues after update	Mid-execution correction
Unrequested continuation	Grasp followed by unwanted placement	Termination and holding

Figure 7 compares $\pi_{0.5}$ and SmartVLA on four diagnostic tasks, highlighting whether execution follows the instructed object–target relation, visual prompt, updated destination, or holding requirement.

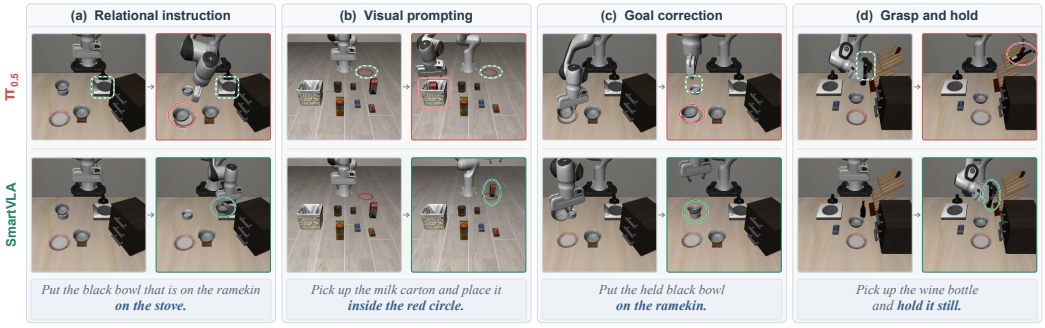


Figure 7: Qualitative comparison of $\pi_{0.5}$ (upper row) and SmartVLA (lower row) under the same diagnostic instructions. Instructions appear below each column; paired frames progress from left to right. (a) Relational instruction: placing the specified bowl on the stove rather than the plate. (b) Visual prompting: placing the milk inside the red circle rather than the basket. (c) Goal correction: redirecting the held bowl to the ramekin rather than the plate. (d) Grasp and hold: keeping the wine bottle grasped rather than continuing to place it on the rack. $\pi_{0.5}$ continues toward an unintended placement target, whereas SmartVLA follows the requested destination or holding condition.

C Representation Probe Diagnosis Protocol

C.1 Probe Setup and Paired Interventions

The probes test whether failures of instruction following coexist with accessible target information in the action-trained representations. If this information is absent, strengthening its correspondence with actions alone cannot exploit an existing semantic representation. Conversely, reliable decoding motivates supervision that links the available information to the required behavior. The probes measure accessibility in the final frozen policies, rather than changes from their original VLMs or the causal contribution of an individual action head.

We freeze $\pi_{0.5}$, GR00T N1.6, and StarVLA and train a common-form conditional target-point probe over each model's multimodal fusion representation. The probe receives language and visual representations associated with the current command and predicts a two-dimensional target point. Behavior and probe results use the same paired instances.

Sem switches fine-grained commands while holding image and robot state fixed, testing whether the target representation changes with semantics. Vis changes scene layout or moves an explicit visual cue, testing whether the representation binds to the current visual position. Con switches target commands in an action-contrast subset, testing whether the internal target representation updates with the control requirement. Sem uses fine-grained commands from the original task templates, Vis uses explicit visual targets under a unified protocol, and Con uses paired commands in the action-contrast subset.

All three models use matched instance pairs, target annotations, and probe-training budgets; the feature location is selected at the corresponding multimodal fusion layer.

C.2 Representation Metrics

Let $\hat{\mathbf{x}}$ be the predicted target point, $\mathbf{x}$ the current target, and $\mathbf{x}'$ the alternative candidate, expressed in the same evaluation-image coordinate system. The per-instance target-point error is $e = \|\hat{\mathbf{x}} - \mathbf{x}\|_2$ in pixels. Target assignment measures the proportion of predictions satisfying $\|\hat{\mathbf{x}} - \mathbf{x}\|_2 < \|\hat{\mathbf{x}} - \mathbf{x}'\|_2$; its two-choice reference is 50%, and it is distinct from behavioral object selection or task completion. Representation response direction measures agreement between the change in predicted target position and the true target displacement across a paired language or visual intervention.

These metrics measure the readability of target information in frozen representations and its response to interventions. They do not directly measure whether actions follow that information and are therefore interpreted jointly with the behavioral results in Appendix B.4.

C.3 Results and Interpretation

Table 7: Behavioral and representation diagnostics for $\pi_{0.5}$, GR00T N1.6, and StarVLA. Obj., Pos., and Comp. are behavioral rates (%), Err. is target-point error in pixels, Assign. is target assignment (%), and Resp. is representation response direction.

Model	Cond.	Obj.↑	Pos.↑	Comp.↑	Err.↓	Assign.↑	Resp.↑
$\pi_{0.5}$	ORI	100.00	100.00	99.00	–	–	–
$\pi_{0.5}$	Sem	88.00	11.00	6.00	14.19	96.63	0.927
$\pi_{0.5}$	Vis	83.00	14.00	5.00	20.26	91.09	0.873
$\pi_{0.5}$	Con	–	25.00	25.00	12.02	97.50	0.978
GR00T N1.6	ORI	100.00	100.00	98.00	–	–	–
GR00T N1.6	Sem	79.00	5.00	1.00	12.26	96.72	0.967
GR00T N1.6	Vis	80.00	3.00	2.00	18.35	92.11	0.896
GR00T N1.6	Con	–	13.00	11.00	16.18	93.89	0.912
StarVLA	ORI	100.00	100.00	96.00	–	–	–
StarVLA	Sem	86.00	3.00	2.00	11.93	98.82	0.981
StarVLA	Vis	72.00	9.00	4.00	23.14	83.80	0.832
StarVLA	Con	–	6.00	6.00	19.66	89.64	0.877

Table 7 reports the paired behavioral and representation diagnostics. Target assignment rates are 96.63–98.82% under Sem, 83.80–92.11% under Vis, and 89.64–97.50% under Con, all above the 50% two-choice reference. The corresponding response-direction scores are 0.927–0.981, 0.832–0.896, and 0.877–0.978. These results indicate that the frozen representations retain decodable target information that varies with fine-grained commands, explicit visual prompts, and updated control requirements.

However, decodability does not imply that the policy effectively uses this information to generate actions. The low behavioral completion rates, despite positive target-assignment and response-direction results, reveal a gap between target information in the representations and execution that follows the current instruction. This contrast supports action-aligned supervision connecting current instructions and visual prompts to local actions, but does not by itself establish complete instruction understanding or identify a unique cause of execution failure.

D Data Construction and Annotation Details

This appendix details the data construction method in Section 2.1, including data sources, quality filtering and unified representations, action-first segmentation, grade-conditioned annotation, instruction rewriting, and exported data statistics.

D.1 Raw Trajectory Sources

The raw trajectory pool consists of DROID (Khazatsky et al., 2024), MolmoAct (Lee et al., 2025), AgiBot-World (AgiBot-World-Contributors et al., 2025), and RH20T (Fang et al., 2023). DROID contains large-scale single-arm Franka teleoperation data collected across real scenes, with 76K trajectories, 564 scenes, and 86 tasks. MolmoAct contains 10,689 high-quality single-arm Franka trajectories over 93 household and tabletop tasks. AgiBot-World provides million-scale real-robot data, including 1,001,552 trajectories, 217 tasks, and 87 skill categories. RH20T contains more than 110K contact-rich robot manipulation sequences with human-operation videos or demonstrations; its task collection comprises 147 tasks. Table 8 reports the complete public scale of each dataset, not the number retained or segmented by SmartVLA; those counts are reported separately below.

Table 8: Public raw trajectory sources used to construct the training pool.

Dataset	Public raw trajectories / sequences	Tasks	Collection coverage
DROID	76K	86 tasks	564 scenes; single-arm Franka
MolmoAct	10k	93 tasks	home and tabletop; single-arm Franka
AgiBot-World	1,001k	217 tasks	106 scenes; 100 real robots
RH20T	110K	147 tasks	7 configurations across 4 robot types

D.2 Pipeline Overview

The SmartVLA pipeline decomposes the conversion from raw trajectories to trainable steering supervision into seven stages. We first filter and canonicalize data across sources, then use action and visual evidence to determine boundaries, object locations, and spatial cues, and finally generate replaceable instruction rewrites. The complete stage mapping is shown in Table 9.

The pipeline separates trajectory analysis, instruction generation, and consistency review. Action and visual evidence first define segment records containing control phases, object roles, and spatial candidates. Grade-conditioned rules then restrict instruction generation to descriptions appropriate for those records, and the review stage removes inconsistent language or spatial labels. This organization reuses existing demonstrations and grounds language generation in structured evidence, rather than requiring unconstrained descriptions to determine the full temporal structure.

Figure 8 illustrates these operations using an RH20T demonstration of slicing lotus root with a knife. The segment-level examples show how one task yields distinct local requirements: approaching and grasping the knife, transferring and aligning it, slicing, and withdrawing. Each segment receives applicable language rewrites and visual prompts tied to its behavior and target, rather than inheriting the same complete-task description. High-interaction segments retain task and visual-prompt rewrites without object-free atomic-motion variants.

Table 9: Seven-stage steering-oriented data construction pipeline.

Stage	Main evidence	Operation	Output
A. Filtering and normalization	language labels, video decodability, image quality, state/action statistics	remove unlabeled, low-quality, or noisy samples and unify image, state, and action formats	valid trajectories in a canonical format
B. Candidate boundary generation	action increments, action curvature, gripper events, visual changes, detection and segmentation	extract candidate peaks from action/visual curves and merge overly short segments by temporal smoothing	candidate control segments

Stage	Main evidence	Operation	Output
C. Interaction evidence modeling	object masks and tracks, gripper-to-target distance, contact evidence, object motion, gripper state	infer active object, target region, contact relation, and control relevance	interaction curves and spatial candidates
D. Action-first refinement	intermediate records from B/C, task-family priors, action segments	refine boundaries, assign a segment grade, and bind semantic actions to object relations	action-aligned segment records
E. Grade-conditioned rewriting	segment grade, original task, action primitive, object/target roles, visual/spatial evidence	select applicable rewrite families by grade and generate candidates	grade-organized rewrite pool
F. Rewrite sampling metadata	task type, phase, rewrite family, segment grade	record recommended sampling weights and applicability conditions	typed rewrite pool with sampling metadata
G. VLM review and expansion	keyframes, rewrite candidates, spatial candidates	review spatial cues and language rewrites, add constrained expressions, and disable unreliable labels	reviewed segment annotations

This decomposition exposes auditable intermediate products. Incoming trajectories pass language, image, action-quality, and canonical-schema checks; boundaries come from action and visual curves; spatial cues come from detection, segmentation, tracking, motion, and contact evidence; and rewrites come from constrained segment records. The final output is not a single caption but two complementary forms of supervision: one describes phase sub-tasks, atomic motions, object roles, and relational constraints, while the other encodes visible targets, regions, and motion trajectories as multimodal inputs. Every training example can be traced to a source segment, interaction evidence, and rewrite type.

D.3 Filtering and Canonical Normalization

We first perform quality filtering. We remove three categories of unlearnable or highly noisy samples. First, episodes or spans without language labels or with empty task text cannot provide a stable instruction anchor. Second, we remove poor visual data, including undecodable videos, black or corrupted frames, severe blur, missing key views, camera jumps, and targets that remain invisible for an extended period. Third, we remove poor actions, including missing state/action values, NaNs or out-of-range actions, long idle periods, clear video-action desynchronization, failed initialization, abnormal gripper state, and trajectories that are too short.

After filtering, action-first segmentation processes only trajectories with readable task language, usable visual observations, and learnable action signals. We map each source to a common multimodal control format. Every sample contains a third-person view and the wrist view of the currently active arm: single-arm trajectories retain one wrist view, while bimanual trajectories retain left and right wrist views separately; side and auxiliary views are retained when available.

To support both single-arm and bimanual embodiments, we use fixed 32-dimensional state and action containers and record embodiment type, active arm, and a valid-dimension mask for every sample. Single-arm data occupies one arm slot. Bimanual data such as AgiBot-World writes the left and right arm state/action into independent slots. Each arm slot primarily represents end-effector pose, gripper state, and relative end-effector action. Base, waist, joint, force, and other information is stored in reserved dimensions or retained as auxiliary proprioception. Padding dimensions unused by the current embodiment do not contribute to the training loss.

Action normalization is also performed at this stage. We prioritize available end-effector poses, Cartesian actions, or end-effector poses reliably computed by forward kinematics. Absolute end-

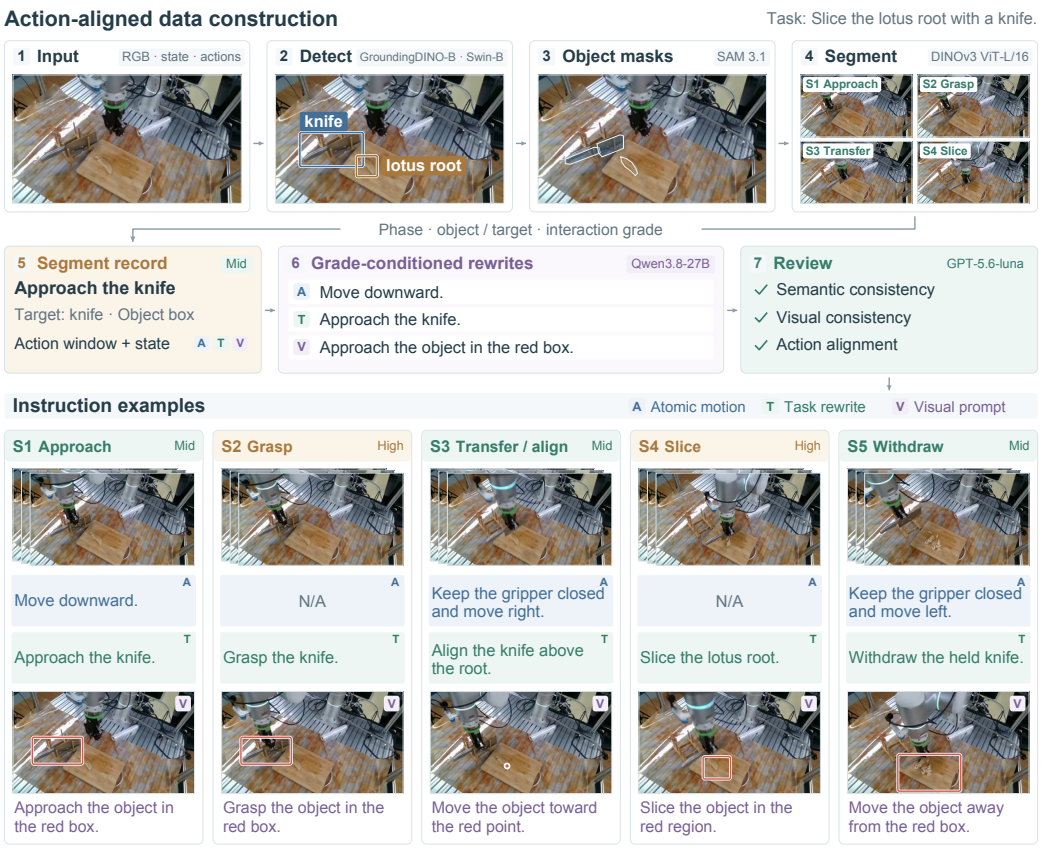


Figure 8: Action-aligned data construction illustrated with an RH20T knife-and-lotus-root demonstration. Top: visual evidence extraction, action-aligned segmentation, segment recording, grade-conditioned rewriting, and consistency review. Bottom: atomic-motion (A), task-language (T), and visual-prompt (V) examples for approach, grasp, transfer/alignment, slicing, and withdrawal. N/A indicates that atomic-motion rewriting is excluded for High-interaction segments.

effector targets are converted into increments relative to the current state, while native end-effector increments retain their control semantics and are reordered into a common axis convention. Normalization statistics are computed per data source and embodiment configuration over valid state and action dimensions, including means, standard deviations, quantiles, and clipping ranges. Different sources therefore share tensor interfaces and dimension semantics while retaining the true scale and control distribution of each robot.

D.4 Action-First Segmentation

SmartVLA derives segment boundaries from action and visual evidence rather than free-form descriptions from a language model. We compute adjacent action-vector differences, action curvature, gripper opening/closing events, embedding shifts in third-person and wrist views, and changes in open-vocabulary object observations. Candidate boundaries are merged into segments using temporal smoothing and task-family rules. This aligns each action segment with a local control phase, rather than binding an episode-level caption to every timestep.

We use DINOv3 ViT-L/16 to compute cross-frame visual changes, GroundingDINO-B with a Swin-B backbone to localize candidate objects in keyframes, and SAM 3.1 to obtain object or target-region masks. Point/box trackers and temporal consistency checks propagate locations through the trajectory. Because the detector requires text queries, we extract names of manipulated and target objects from the original instruction and augment them with descriptions implied by the candidate

segment phase. We then combine gripper-to-target distance, contact evidence, object motion, and gripper state to infer the active object, target region, and interaction phase for each segment.

The resulting action segments provide local steering supervision, while original-task examples retain their continuous demonstration trajectories. Unused embodiment dimensions remain masked as described above.

D.5 Grade-Conditioned Steering Instruction Rewriting

Each record stores the action segment, its control phase, interaction grade, and available object and spatial evidence. The grade selects among the three rewrite families defined in Section 2.1, rather than filtering segments. Table 10 associates typical phases with the permitted families.

Table 10: Grade-conditioned instruction rewrite applicability.

Grade	Typical phase	Object-free atomic motion	Synonym/task rewrite	Spatial visual generalization
Low	free-space motion, reset, recovery, generic movement	yes	yes	no
Mid	reach, align, transport, approach, retreat	yes	yes	yes
High	grasp, place, insert, press, open/close contact	no	yes	yes

Low segments use atomic-motion and synonym/task rewrites without spatial labels, avoiding unnecessary spatial references when no clear interaction target exists. Mid segments use all three families to connect motion direction and phase intent with visible targets. High segments use synonym/task rewrites and spatial visual generalization, preserving object, contact, and local-target information rather than simplifying explicit interactions into object-free motion.

We use Qwen3.8-27B (Qwen Team, 2026) to generate candidate instructions under the rewrite families permitted by each grade, based on segment records and annotation rules. We use GPT-5.6-luna to review these candidates for semantic and visual consistency and remove unreliable language or spatial labels. The same action segment can therefore provide diverse supervision at an interaction-appropriate granularity while preserving consistency among instruction meaning, visual evidence, and action segment.

D.6 Data Statistics

Table 11: Statistics of the four source-based SmartVLA training subsets. Mean length is measured in canonical sampled frames.

Training subset	Included traj.	Mean frames	Segments
DROID	74,731	265	446,892
MolmoAct	7,817	141	49,559
RH20T	12,495	507	77,719
AgiBot-World	50,136	321	284,745
Total	145,179	—	858,915

Table 11 summarizes the included trajectories, their mean lengths, and the total number of action segments. Trajectory length is measured in canonical sampled frames after source-specific preprocessing, describing data scale rather than comparable physical duration across sources. Each retained action segment is counted once, irrespective of its number of rewrites or reuse across training stages; repeated training-time sampling does not increase the segment count.

Figure 9 visualizes the source composition, action vocabulary, and phase-level statistics of the constructed corpus.

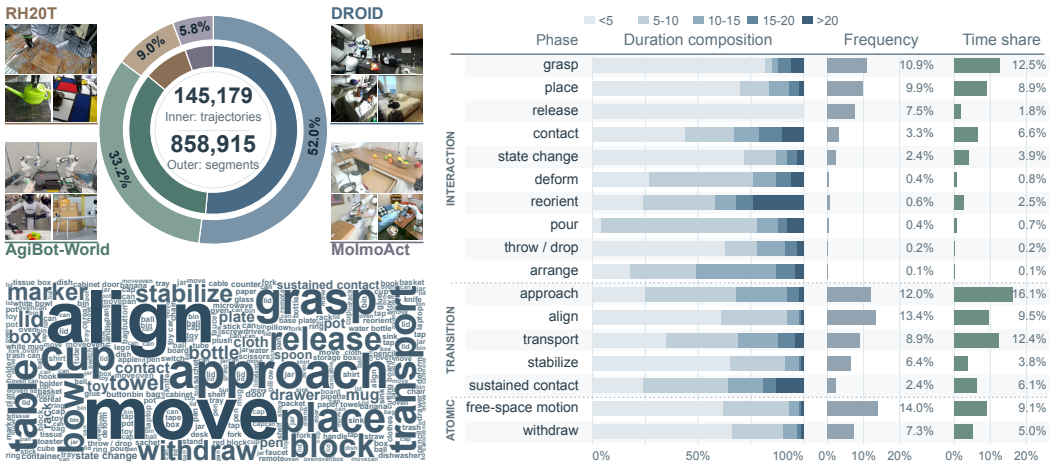


Figure 9: SmartVLA data construction. Left: four trajectory sources, aligned segments, and the action vocabulary word cloud. Right: subtask duration composition, frequency, and total time across interaction, transition, and atomic-motion phases.

E SmartVLA-Pi and SmartVLA-OFT: Architecture, Training, and Calling Protocol

This appendix describes the two policy instantiations used in the experiments. SmartVLA changes the supervision and training procedure, while retaining the native action models of the two base VLA families. Both variants consume the same type of caller-provided instruction, third-person and wrist RGB observations, and robot state, and return a finite local action block. The regression and flow-matching losses in Equations (5) and (7) supply the native losses used in the shared training formulation of Equation (1).

E.1 Common Instruction-Conditioned Interface

At time t, each policy maps the current multimodal observation and command to a local action distribution,

$$\mathbf{a}_{t:t+H-1} \sim \pi_\theta(\cdot \mid \mathbf{o}_t, \mathbf{s}_t, c_t), \tag{2}$$

where c_t can be a complete task instruction, a phase-level subgoal, an atomic motion, a spatially grounded command, or a corrective instruction. The visual markers used by LIBERO-SMART are rendered in the RGB inputs and are not supplied through privileged channels. The interface in Equation (2) is used for end-to-end task execution and for repeated calls from an upper-level controller.

E.2 SmartVLA-OFT

SmartVLA-OFT uses Qwen3-VL-4B-Instruct (Bai et al., 2025) as its vision-language backbone and an OFT-style residual MLP for continuous action regression. The backbone combines a 24-layer vision transformer with a 36-layer language transformer of hidden width 2,560. Multi-view image tokens, the current instruction, and a textual encoding of normalized robot state form the conditioning sequence. We append H action-query tokens, one for each predicted step. The final-layer hidden states at these positions are extracted together in a single forward pass:

$$\mathbf{Z}_t = \mathrm{Gather}_{\mathrm{act}}[F_\psi(\mathbf{o}_t, c_t, \mathrm{Tok}_s(\mathbf{s}_t), \mathbf{q}_{1:H})] \in \mathbb{R}^{H\times 2560}. \tag{3}$$

In Equation (3), Tok_s denotes state discretization and textual encoding, and $\mathbf{q}_{1:H}$ are fixed action placeholders rather than previously generated actions. A shared MLP maps each query state to

a normalized continuous action. It uses a 5,120-dimensional hidden representation, two residual blocks with LayerNorm and ReLU, and a 32-dimensional output projection:

$$\widehat{\mathbf{A}}_t = g_\phi(\mathbf{Z}_t), \qquad \mathbf{A}_t^{\text{exec}} = \mathcal{N}^{-1}\Big(\widehat{\mathbf{A}}_t[:, \mathcal{I}]\Big). \tag{4}$$

In Equation (4), the index set $\mathcal{I}$ selects the active robot action dimensions, and $\mathcal{N}^{-1}$ applies the corresponding action denormalization. Training uses an L_1 loss on these active dimensions:

$$\mathcal{L}_{\text{OFT}} = \frac{1}{H|\mathcal{I}|} \sum_{k=0}^{H-1} \sum_{j \in \mathcal{I}} \left| \widehat{A}_{t,k,j} - A_{t,k,j} \right|. \tag{5}$$

The LIBERO instantiation uses two 256×256 RGB views, an eight-dimensional state encoded on a 0–1000 scale, $H = 8$, and seven active action dimensions. The whole chunk is predicted at once, without autoregressive action-token decoding or iterative flow sampling. Action-aligned segments and grade-conditioned rewrites provide the semantic and spatial supervision for this otherwise unchanged continuous-regression interface.

E.3 SmartVLA-Pi

SmartVLA-Pi uses the $\pi_{0.5}$ architecture implemented in OpenPI: a PaliGemma vision-language backbone comprising a SigLIP-So400m/14 visual encoder and a Gemma-2B language model, paired with a separate approximately 300M-parameter action expert. The expert has 18 transformer layers and hidden width 1,024. Images, language, and discretized robot state form the multimodal prefix. In this $\pi_{0.5}$ configuration, state is encoded with the language tokens rather than injected as a separate continuous state token into the action suffix. Noisy actions are projected into expert tokens, while a sinusoidal flow-time embedding and an MLP condition the expert through adaptive RMS normalization. Expert tokens attend to the multimodal prefix and to one another.

Let $\mathbf{A}_t \in \mathbb{R}^{H \times 32}$ denote the normalized demonstration action chunk and $\epsilon \sim \mathcal{N}(0, I)$ Gaussian noise. Using the convention that $\tau = 1$ is noise and $\tau = 0$ is the action target, training constructs

$$\mathbf{X}_t^\tau = (1 - \tau)\mathbf{A}_t + \tau\epsilon, \qquad \mathbf{u}_t = \epsilon - \mathbf{A}_t. \tag{6}$$

Let $\mathcal{I}$ denote the active action dimensions of the current embodiment and $\mathbf{M}$ the binary mask retaining those dimensions at every predicted step. Using the interpolation and velocity target in Equation (6), the conditional velocity field v_θ is learned through a flow-matching loss evaluated only on active dimensions,

$$\mathcal{L}_{\text{Pi}} = \mathbb{E}_{\mathbf{A}_t, \epsilon, \tau}\left[\frac{1}{H|\mathcal{I}|} \left\| \mathbf{M} \odot \left(v_\theta(\mathbf{X}_t^\tau, \tau; \mathbf{P}_t) - (\epsilon - \mathbf{A}_t) \right) \right\|_F^2 \right], \tag{7}$$

where $\mathbf{P}_t$ denotes the encoded multimodal prefix. Padded dimensions are excluded from the loss in warmup, pretraining, and adaptation alike. At inference, the prefix is encoded once and its attention keys and values are cached. Starting from $\mathbf{X}_t^{(0)} = \mathbf{M} \odot \epsilon$, the action expert performs N Euler steps from noise to actions:

$$\begin{aligned} \tau_n &= 1 - \frac{n}{N}, \qquad n = 0, \ldots, N-1, \\ \mathbf{X}_t^{(n+1)} &= \mathbf{M} \odot \left[\mathbf{X}_t^{(n)} - \frac{1}{N} v_\theta(\mathbf{X}_t^{(n)}, \tau_n; \mathbf{P}_t) \right], \\ \mathbf{A}_t^{\text{exec}} &= \mathcal{N}^{-1}\Big(\mathbf{X}_t^{(N)}[:, \mathcal{I}] \Big). \end{aligned} \tag{8}$$

The LIBERO instantiation uses 224×224 model-input images, $H = 10$, $N = 10$, and seven active dimensions in the 32-dimensional action container; inactive dimensions remain zero during sampling. In Equation (8), N counts internal integration steps, whereas H counts predicted environment actions. Prefix caching avoids rerunning the vision-language backbone at every integration step. SmartVLA changes the instruction and segment-level action supervision while retaining this flow-matching action generator.

E.4 THREE-STAGE OPTIMIZATION

Both variants follow the same three-stage schedule: warmup, pretraining, and adaptation. Warmup uses object-free atomic-motion commands and local action segments. With images omitted and robot state retained, the entire VLM backbone is frozen while the action head learns the correspondence between a short command and its local control. Pretraining restores the multimodal observations and jointly trains the policy on original instructions and grade-permitted semantic and spatial rewrites. Adaptation then fine-tunes the complete policy independently for each target domain, mixing original task commands with the applicable steering supervision. Original-task examples preserve the continuous episode trajectory, whereas local examples use the corresponding action-aligned segment. This separates complete-task imitation from local command following while keeping both forms accessible through one instruction interface.

We collectively refer to the checkpoints obtained after warmup followed by pretraining as **SmartVLA-Base**, with two instantiations: **SmartVLA-Pi-Base** and **SmartVLA-OFT-Base**. The warmup checkpoint is an intermediate initialization for pretraining. For each architecture, its Base checkpoint initializes independent adaptation runs for LIBERO, RoboTwin Full, RoboTwin C2R, and Piper. Each run produces a separate adapted expert; adaptation does not proceed sequentially from one target-domain expert to another. Unless otherwise specified, SmartVLA results use the corresponding adapted experts, reported as SmartVLA-Pi and SmartVLA-OFT. The Base suffix identifies the pretrained checkpoint before target-domain adaptation.

SmartVLA-Pi and SmartVLA-OFT use the same listed training settings and update counts within each stage and target-domain protocol. Shared warmup and pretraining settings are reported once in Table 12, with stage-specific learning rates indicated explicitly. Adaptation settings are listed by target domain in Table 13. Training steps count optimizer updates, and the effective global batch size includes data parallelism and gradient accumulation.

Table 12: Shared warmup and pretraining hyperparameters for SmartVLA-Pi and SmartVLA-OFT. Stage-specific values are indicated explicitly.

Parameter	Shared configuration
Trainable modules during warmup	Action head; VLM backbone frozen
Trainable modules during pretraining	All components; no freezing
Original/local instruction sampling (pretraining and adaptation)	1:1 effective probability; not applicable to warmup
Hardware	16× NVIDIA A100
Effective global batch size	256
Optimizer	Fused AdamW
Peak learning rate (warmup / pretraining)	5×10^{-5} / 10^{-5}
Adam coefficients (β_1, β_2)	$(0.9, 0.95)$
Optimizer epsilon	10^{-8}
Weight decay	0.1
Learning-rate schedule	`cosine_with_min_lr`
Learning-rate warmup ratio	0.01
Gradient-clipping threshold	1.0
TF32	Enabled
Parallelism	DeepSpeed ZeRO-2
State normalization (`state_mode`)	`QUANTILE` (Q01–Q99)
Training random seed	42
Optimizer updates (warmup / pretraining)	10,000 / 240,000

Table 13: Shared adaptation hyperparameters for SmartVLA-Pi and SmartVLA-OFT, listed by target domain and protocol. Seed denotes the training random seed.

Domain / protocol	Updates	Global batch	Peak LR	Seed
LIBERO	50,000	256	10^{-5}	42
RoboTwin Full	120,000	256	10^{-5}	42
RoboTwin C2R	50,000	256	10^{-5}	42
Piper	50,000	64	10^{-5}	1000

E.5 Agentic Calling Protocol

In Table 2, direct-policy results use all 6,000 instances (120 tasks, 10 macro layouts, and 5 local initializations per layout). The agentic protocol uses the same task contracts, and model comparisons are made within each protocol.

In an agentic system, one upper-level call specifies a natural-language command and the number of action blocks to execute. A single low-level VLA prediction is $\mathbf{a}_{t:t+H-1} \sim \pi_\theta(\cdot \mid \mathbf{o}_t, \mathbf{s}_t, \mathbf{c}_t)$, where $\mathbf{o}_t$ contains the current third-person and wrist RGB observations. Colors, regions, trajectories, and target points involved in the task appear directly in these images rather than through privileged channels. $\mathbf{s}_t$ denotes end-effector pose and gripper state, and $\mathbf{c}_t$ is the natural-language command supplied by the caller. It may be a complete task, a current subgoal, or a local corrective command based on execution. SmartVLA predicts a local block of length H; our Pi-based agentic evaluations use $H = 10$. This is the length of one action block, not the total number of environment steps in an upper-level call.

Each upper-level call also specifies the number K of action blocks, corresponding to a budget of KH environment action steps. Within the call, the VLA executes blocks under the same command: after each block, the system obtains updated observations and robot state and predicts the next block. After K blocks, control returns to the caller. Thus, one upper-level call can contain multiple action blocks and still use updated observations for closed-loop control; it is not required to complete an entire command or long-horizon task.

The human operator or upper-level agent can use the returned observation to call the VLA again with the original command, a sub-task command, or a corrective command, and can choose a new block count K. It may instead call an atomic skill in the agentic system or terminate the episode. Both K and the number of calls are selected dynamically from execution progress and remain subject to a common interaction budget.

For the Agentic+$\pi_{0.5}$ and Agentic+SmartVLA-Pi comparison, we use GPT-5.6 as the fixed upper-level planning model. We fix the upper-level controller, available tools, calling protocol, and re-planning rules, and keep tasks, initializations, and interaction budgets identical while replacing only the low-level VLA. The upper controller decides from observations when to call the VLA, which command to use, and how many action blocks to execute, but cannot access environment rewards, online success signals, or hidden evaluation state.

CaP-X uses its own harness and is reported as a separate complete-system baseline rather than as a controlled replacement of the low-level VLA.

The agentic systems use the same scoring rules. An episode terminates when the environment-step budget is exhausted or the upper-level VLM reports task completion, whichever occurs first. A completion report is a stopping decision, not an evaluator-provided success signal. The evaluator retains the episode's contract history across VLA calls and tool invocations; returning control to the upper level does not clear forbidden events or the recorded step sequence. Success is determined by strict contract satisfaction at termination, without the additional 30-step success window used for end-to-end policies. A VLM completion report alone does not count as success.

All agentic scores measure **overall Agent+VLA performance**, not the standalone capability of the VLA. The fixed-controller comparison between Agentic+$\pi_{0.5}$ and Agentic+SmartVLA-Pi isolates the effect of replacing the low-level policy within that shared upper-level system.

F BENCHMARK EVALUATION SETTINGS AND LIBERO-SMART PROTOCOL

Benchmark adaptation and evaluation convention. Our cross-model comparisons evaluate the downstream capabilities of trained policies. Models retain their respective pretraining data, objectives, and training budgets. Except where explicitly noted, baselines follow the standard training or adaptation protocol released with each benchmark: they are either SFT-adapted on the corresponding training data or evaluated using a publicly released target-domain adapted checkpoint. For evaluations conducted in this work, we match tasks, initializations, interaction horizons, and success criteria within each benchmark and evaluation protocol. Test-time evaluation follows the native benchmark protocol; LIBERO-SMART uses the protocol specified below. Its evaluation instances are not used to adapt any model. The scope of the within-model component ablations is described in Appendix H.

LIBERO, LIBERO-Plus, and LIBERO-Pro. These benchmarks evaluate basic task execution, perturbation robustness, and task and position generalization, respectively. In our evaluations, we retain their native goal-based success criteria and interaction horizons, using matched tasks and initializations across methods within each benchmark.

For each SmartVLA architecture, the same LIBERO-adapted expert is evaluated on LIBERO, LIBERO-Plus, LIBERO-Pro, and LIBERO-SMART, without further benchmark-specific adaptation. LIBERO-SMART evaluation instances are not used in any of these adaptation runs.

RoboTwin2.0. Full adapts on clean and randomized demonstrations and evaluates both scene types; Clean2Random (C2R) adapts only on clean demonstrations and evaluates clean and randomized scenes. The two protocols use independently adapted experts.

LIBERO-SMART. We evaluate the three instruction-following capabilities using the trajectory-level task contracts in Section 3, with matched instances and interaction budgets across models within each evaluation protocol. The agentic calling protocol and success aggregation rules are detailed in Appendices E.5 and F.3.

Piper and training. Piper evaluates instruction-following gains in real-world bimanual control. SmartVLA uses a separate expert for each target domain following Section 2.2; model and execution details are provided in the appendices.

F.1 TASKS AND INITIALIZATION INSTANCES

The benchmark comprises three independently scored suites. The task and initialization counts are summarized in Table 14.

Table 14: LIBERO-SMART task suites and initialization counts.

Suite	Tasks	Init./task	Instances	Core content
Semantic	40	50	2,000	paraphrases, sub-tasks, atomic motions, constraints
Visual Prompt (Vis)	40	50	2,000	object circles, regions, points, trajectories
Control	40	50	2,000	target correction and process re-planning
Total	**120**	**50**	**6,000**	three capabilities

Every task has 10 macro layouts and 5 local initialization variants per layout. For language-reference tasks, the target relation must hold in the current scene with a unique referent; left/right relations are resolved from the task-specified view rather than a fixed object identifier. For visual-prompt tasks, cue position, target entity, and spatial predicate are updated together for each instance. For Control, the expected task phase and required held or completed state are preserved while scene placement and feasible robot pose are varied.

Controlled initialization for corrective instruction following. For each Control instance, all models begin from the same intermediate physical state and are evaluated on their response to the cor-

rective instruction. This fixes the robot configuration, object arrangement, and required interaction state at the start of the evaluated continuation. If each model instead executed the preceding stage independently, differences in grasp pose, object placement, or incomplete preceding actions could change the difficulty of the correction. Standardized initialization controls these preceding-stage effects and permits comparison of how models execute the updated requirement from the same physical starting point. Models receive the same corrective instruction and interaction budget within each evaluation protocol.

Validity checks cover key-object visibility, target reachability, support and contact relations, and object stability. We exclude geometric intersections, unreasonable occlusion, or extra obstacles that change instruction meaning. Tasks requiring a new operation exclude initial states in which the target is already complete. Holding tasks retain the required initial state and test whether the policy maintains it during execution. Initial states are specified reproducibly and shared across models.

Each Visual Prompt task fixes its cue type and color semantics, while initializations vary the object or position referred to by the cue. Cue colors are selected according to scene contrast. The markers modify only the policy's image inputs and do not alter physical collision relations.

Before formal comparisons, representative trajectories are manually reviewed to verify that object identity, process constraints, termination, and stability clauses are consistent with the task semantics. Physically invalid initializations or semantically ambiguous instances are repaired or excluded before the final aggregate is computed.

F.2 Per-Step Trajectory-Contract Satisfaction

LIBERO-SMART represents requirements as a trajectory-level contract that jointly captures target correctness, process consistency, and result stability. For task i, let $\tau_{\leq t}$ denote the execution history through step t. Strict contract satisfaction at that step is

$$C_i(t) = \mathbf{1}[G_i(s_t) \wedge H_i(\tau_{\leq t}) \wedge R_i(\tau_{\leq t})]\,, \tag{9}$$

where G_i requires the current result to satisfy the current instruction, H_i requires applicable object restrictions, prohibitions, and step order throughout the observed history, and R_i requires a semantically valid and stable result. These terms answer "is the result correct?", "was the instruction followed?", and "was the task actually completed?", rather than treating a single transient target state as complete instruction following.

Task goals and applicable prohibitions are evaluated from environment state and predicates. Requirements involving execution history are checked from the trajectory, including forbidden events, state changes of protected objects, wrong-object manipulation, and the order of multi-step events. Tasks without a corresponding language constraint do not receive an additional restriction.

Termination conditions are task-specific. Placement requires release and stable support; grasping or holding requires the correct grasp and corresponding stable state; switching tasks check device state and instruction. Stability checks require applicable state conditions to hold for the number of consecutive steps specified by the task. Trajectory tasks use more permissive criteria for trajectory matching, release, and stability and do not require pointwise replay of the drawn curve.

The evaluation interface receives actions generated from images, robot state, and the current instruction. Task binding and contract checking are performed by an independent evaluator. Additional simulator information used to check target and trajectory constraints is not provided as policy input.

A forbidden event remains part of the observed history even if the final state is later repaired. For example, placing a non-target object in a basket and removing it still violates an instruction that forbids placing other objects there. Step-order requirements likewise depend on the recorded event sequence, not only on the set of completed operations.

F.3 Interaction Horizon and Success Aggregation

End-to-end evaluation requires attaining strict success within the original interaction horizon and maintaining the task contract for 30 full subsequent environment action steps, rather than ending the check at a single successful state. The policy continues to execute actions conditioned on current observations during the holding check.

Let $C_i(t)$ denote strict contract satisfaction at step t, as defined in Equation (9), and let T_i denote the original interaction horizon. Let $\mathcal{T}_i$ be the set of environment steps in the task's valid evaluation interval that do not exceed T_i. The success indicator is

$$S_i^{30} = \mathbf{1}\left[\exists\, t \in \mathcal{T}_i : \quad \prod_{k=0}^{30} C_i(t+k) = 1\right]. \tag{10}$$

The holding window in Equation (10) may extend beyond the original horizon, but must start from a successful state at or before T_i and cover all 30 subsequent steps; evaluation therefore extends at most to $T_i + 30$. Extra steps only complete a holding check and cannot initiate a new success window beyond the horizon.

If a candidate window is interrupted without violating history-dependent constraints, the policy may attain success again and start another window within the original horizon. Object restrictions, forbidden events, and step-order history persist throughout the episode and are not reset by a retry. Success requires every step in a window to satisfy the same strict contract, including applicable history-dependent constraints and task-specific stability requirements. If no such window exists, the episode fails. Ever-success and strict terminal success are distinct auxiliary statistics, not substitutes for this indicator.

For agentic systems, let t_i^{end} be the environment step at which the budget is exhausted or the upper-level VLM reports completion, whichever occurs first. The success indicator is $S_i^{\mathrm{agent}} = C_i(t_i^{\mathrm{end}})$, evaluated independently at termination. No additional 30-step window is imposed. The contract's task-specific stability and release or holding conditions still apply at termination, and history-dependent constraints are checked over the entire episode. Neither an earlier successful state nor the VLM's completion report alone establishes success. All three agentic systems use this same rule.

As an auxiliary metric, Goal-only records whether the current SMART instruction's positive goals are simultaneously satisfied at any post-action state within the episode's original step budget and before termination. It preserves the specified objects, targets, and instantaneous spatial relations, but omits forbidden-event history, action-order constraints, displacement-preservation guards, release checks, velocity thresholds, and stability or holding windows. It is distinct from both ever-strict success and success on the original LIBERO task. The column averages this indicator over the three suites.

Results are first averaged over initialization instances for each task, then aggregated with equal task weight within each suite and equal suite weight across the three suites. Within each evaluation protocol, models use identical tasks, initializations, interaction budgets, and task contracts; the end-to-end and agentic evaluation settings are specified in Appendix E.5.

F.4 Relationship Between Training Data and Evaluation Tasks

During LIBERO target-domain adaptation, supervision is derived from the original LIBERO demonstrations. We perform action-aligned segmentation and generate local instructions, linguistic rewrites, and spatial visual supervision from the corresponding observations and action segments; execution trajectories of LIBERO-SMART evaluation tasks are not used as training demonstrations. The additional supervision therefore expands the instruction expressions and visual conditions associated with existing manipulation experience, rather than introducing demonstrations collected for the evaluation tasks.

LIBERO-SMART shares LIBERO's basic manipulation skills and scene assets, but its evaluation instances do not reuse the corresponding training trajectories. The benchmark changes the conditions under which familiar skills are invoked through held-out task combinations, language conditions, object–target relations, and macro scene layouts. Basic primitives such as grasping, lifting, and placement may therefore be shared, while their phase-specific combinations and semantic bindings are new at evaluation time.

The exact evaluation instructions are new instantiations rather than copies of the original episode captions. They combine object roles, target relations, ordering, exclusion, and phase-specific termination requirements that are not present in the corresponding training episodes. Likewise, evaluation

visual cues use varied colors and locations, and their association with the current object or target is changed across instances. Circles, points, regions, and trajectories are thus not tied to a fixed coordinate or a fixed object identity.

This separation preserves the controllability provided by familiar objects and basic skills while preventing a fixed appearance or object–location association from being a sufficient solution. The benchmark consequently measures whether the policy can apply learned instruction-following patterns to new semantic, relational, and visual configurations, rather than directly imitating evaluation trajectories.

G Detailed LIBERO and LIBERO-Plus Results

Tables 15 and 16 provide the suite and perturbation breakdowns of the aggregate scores in Table 3.

G.1 LIBERO

Table 15: LIBERO benchmark success rates (%).

Method	Spatial	Object	Goal	LIBERO-10	Average
OpenVLA	84.70	88.40	79.20	53.70	76.50
OpenVLA-OFT	97.60	98.40	97.90	94.50	97.10
π_0	96.80	98.80	95.80	85.20	94.15
$\pi_{0.5}$	98.80	98.20	98.00	92.40	96.85
X-VLA	98.20	98.60	97.80	**97.60**	98.05
GR00T N1.6	97.50	98.50	97.50	94.40	96.98
StarVLA	97.80	98.60	96.20	93.80	96.60
APT	91.60	70.00	84.60	42.80	72.25
SmartVLA-Pi	**99.00**	**99.40**	98.00	96.80	**98.30**
SmartVLA-OFT	98.80	99.00	**99.20**	95.80	98.20

G.2 LIBERO-Plus

Table 16: LIBERO-Plus success rates (%). Overall is the episode-weighted aggregate over the seven perturbation categories.

Method	Camera	Robot	Language	Light	Bg.	Noise	Layout	Overall
OpenVLA	0.80	3.50	23.00	8.10	50.40	15.20	28.50	17.28
OpenVLA-OFT	56.40	31.90	79.50	88.70	97.30	75.80	74.20	70.02
π_0	13.80	6.00	58.80	85.00	90.70	79.00	68.90	54.63
$\pi_{0.5}$	64.80	71.80	83.00	93.50	92.20	78.80	**85.50**	80.26
X-VLA	23.40	**89.70**	75.70	88.20	96.00	62.70	71.80	70.46
StarVLA	48.70	63.40	86.80	95.80	94.60	75.00	80.20	76.08
ABot-M0	60.40	67.90	86.40	**96.20**	91.60	86.40	82.60	80.49
APT	16.57	44.58	59.47	76.44	78.72	63.52	65.25	55.85
SmartVLA-Pi	**68.70**	74.70	87.20	94.50	**97.90**	**89.90**	82.40	**84.00**
SmartVLA-OFT	66.30	72.50	**88.50**	96.10	96.70	83.10	84.50	82.76

H Ablation Protocol Details

The standard component ablations use SmartVLA-Pi and the same target-domain source trajectories, adaptation update count, random seeds, evaluation instances, and trajectory-level success contract as the full SmartVLA-Pi model. Table 17 defines the settings evaluated in Table 5.

The ablations address three aspects of the training framework. Stage-removal experiments evaluate the contribution of each stage to the complete training procedure, including the training exposure associated with that stage. Data-construction controls examine the choices used to form local supervision: uniform segmentation changes the temporal boundaries, while removing grade conditioning changes the applicability rules for instruction families. The raw-data adaptation control tests whether local instruction supervision remains important during target-domain adaptation. The fine-grained semantic-only setting examines whether enriching whole-trajectory language is sufficient to obtain the instruction-following gains of the full framework. It applies fine-grained rewrites to complete trajectories during pretraining and adaptation, without warmup, action-aligned local segmentation, or visual-prompt supervision. This setting implements a FineVLA-inspired supervision strategy within our experimental pipeline.

Table 17: Definitions of the SmartVLA-Pi ablation settings.

Setting	Removed or changed component
Full SmartVLA-Pi	warmup, pretraining, and adaptation with all steering-oriented supervision
w/o warmup	Skip image-free warmup
w/o pretraining	Skip pretraining and adapt from the warmup checkpoint
From raw VLM	Skip SmartVLA's warmup and pretraining and begin target-domain training from the original VLM weights; retain the full method's adaptation data processing and steering supervision
Raw-data adaptation	Retain the warmup and pretraining weights, but fine-tune directly on the corresponding benchmark's original demonstrations and task instructions, without action-aligned segmentation, local-instruction rewriting, or spatial visual annotation
Uniform segmentation	Replace action-aligned segmentation with fixed-interval temporal cuts, approximately matching the full method's segment count and mean length; retain the subsequent annotation and training procedures
w/o grade conditioning	Retain the three rewrite families but remove the High/Mid/Low rules governing which families are used; apply a common rewriting rule while retaining semantic and visual consistency review
Fine-grained sem. only	Two-stage pretraining and adaptation using fine-grained language rewrites on complete trajectories in both stages; no warmup, action-aligned local segmentation, or visual-prompt supervision

The two data-construction controls each change only the specified component and retain original complete-task supervision and the original-to-local supervision sampling ratio. Uniform temporal segmentation approximately matches the full method's segment count and mean length, with subsequent annotation and training procedures retained. Its average success rate of 7.45%, compared with 14.67% for the full model, supports the importance of aligning segment boundaries with control phases. Removing grade conditioning yields 12.25%, supporting the use of interaction-dependent instruction families.

The fine-grained semantic-only setting reaches 7.00% on average, showing that whole-trajectory language enrichment alone does not recover the full framework's gains in this setting. The temporal segmentation and grade-conditioning controls provide more targeted evidence for the roles of local alignment and rewriting applicability. Together, these comparisons support the design of action-aligned, grade-conditioned supervision for local instruction following.

I Real-Robot Piper Evaluation Details

I.1 Experimental Platform and Configuration

The real-robot experiments use a bimanual Piper setup. Table 18 summarizes the experimental platform, sensing and control settings, and evaluation scale.

Table 18: Piper real-robot sensing, execution, and evaluation configuration.

Parameter	Configuration
Robot platform	Bimanual Piper setup
RGB acquisition resolution	Main view and two wrist-mounted views, each at 640×480
Camera and control frequency	30 FPS per camera, matched to the robot control frequency
Predicted action chunk length	50 control steps
Executed actions per prediction	25 control steps
Task and trial counts	10 tasks per capability, 20 trials per task
Episode budget	1,000 control steps

I.2 Task Composition and Trial Arrangement

Semantic, Visual Prompt, and Control each contain 10 tasks, with 20 trials per task, for a total of 600 trials per model.

The representative trajectories below use one task from each category: folding a towel from bottom to top (Semantic), placing the bowl marked by a red circle on the shelf (Visual Prompt), and changing the Control instruction during execution from placing a sticker on top of a box to placing it on the front.

I.3 Representative Execution Trajectories

Each category is illustrated by one representative execution video. Frames within each contact sheet are ordered from left to right and then from top to bottom, providing a compact summary of the corresponding behavior.

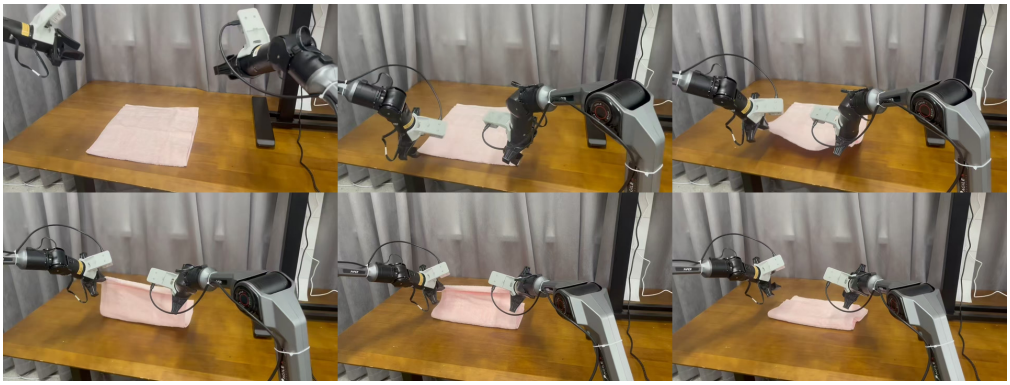


Figure 10: Piper Semantic task execution. Task instruction: “Fold the towel from bottom to top.”

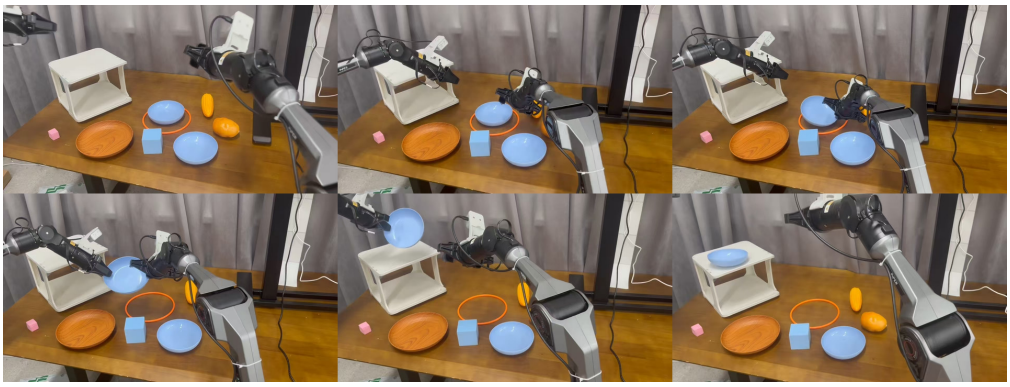


Figure 11: Piper Visual Prompt task execution. Task instruction: “Place the bowl marked by the red circle on the shelf.”

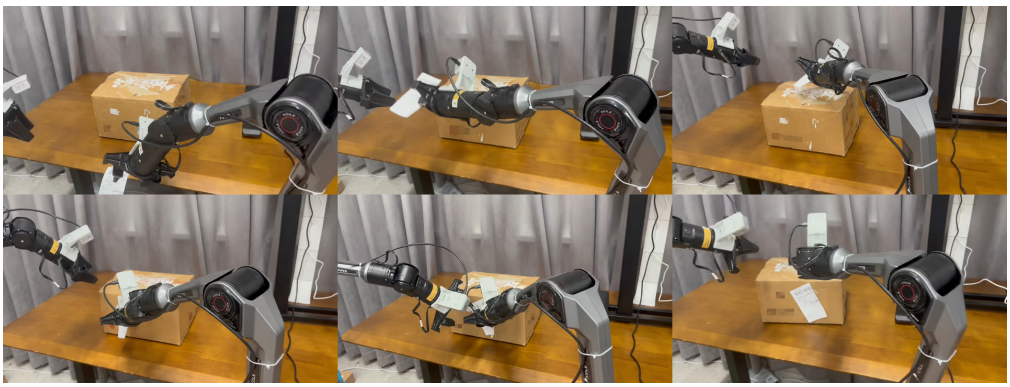


Figure 12: Piper Control task execution. The initial instruction, “Attach the sticker to the top of the box,” is replaced during execution by “Attach the sticker to the front of the box.”

J LIMITATIONS AND FUTURE WORK

This work focuses on enabling existing manipulation skills to be flexibly invoked through diverse instructions, with evaluation covering representative scenarios involving fine-grained semantic following, multimodal instruction understanding, and mid-execution corrective control. More open-ended task spaces, longer-horizon instruction composition, and capability transfer across robot platforms remain directions for further exploration. Future work will broaden the coverage of instructions and interaction scenarios, investigate how instruction-conditioned policies can work with high-level planning and safe execution mechanisms, and continue to develop LIBERO-SMART to support broader research on instruction following and interactive control.